ZAtop Gearless elevator machine

SM200

SM200.15C/20C/30C

SM200.40C

Description

- Permanent magnet synchronous motor with NdFeB magnets
- With ≤ 320 mm installation width for the narrowest shaft dimensions
- High efficiency
- Noise emissions: < 50 dB(A)
- Axle load: up to 3,300 kg
- Traction sheave: 160 mm to 500 mm
- Separately controllable brake circuits, type-tested as safety device against uncontrolled and unintended car movement
- Temperature monitoring by PTC or KTY thermistor

Scope of delivery

- Gearless elevator machine
- Type-tested brake system
- Traction sheave
- Absolute encoder type Heidenhain ECN1313 EnDat
- 5 m Motor cable in SM200.15C to SM200.30C
- Rope guard according to EN 81

Options

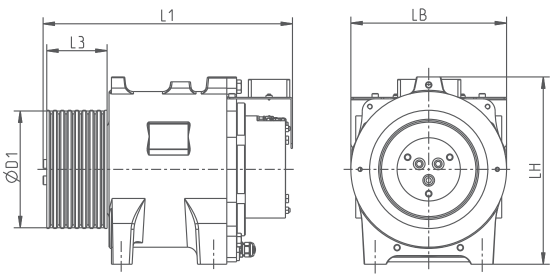
- Absolute encoder type Heidenhain ECN1313 SSI or ERN1387
- Motor cable in lengths 5 m, 10 m, 15 m, 20 m and 25 m
- Mechanical hand release system for brake
- Pre-parametrised frequency inverter ZAdyn

Technical data

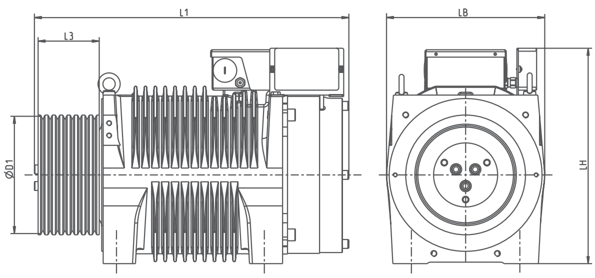
Motor type	Nominal torque	Max. axle load	Nominal speed	Rated output power
	Nm	kg	min^{-1}	kW
SM200.15C	250	1850	96...300	2.5...7.9
SM200.20C	330	2850		3.3...10.4
SM200.30C	475	2850		4.8...14.1
SM200.40C	600	3300		6.0...18.8

Dimensions mm

SM200.15C, SM200.20C, SM200.30C



SM200.40C



Motor type	L1	LH	LB	D1	L3	Weight
	mm	mm	mm	mm	mm	kg
SM200.15C	487	385	320	160	76	167
				210	76	175
				240	88	180
				320	74	185
SM200.20C	517	385	320	160	106	187
				210	106	195
				240	124	205
				320	110	210
				400	92	230
				450	92	235
SM200.30C	562	398	320	210	106	225
				240	124	235
				320	110	240
				400	92	260
				450	92	265
SM200.40C	676	437	320	160	143	295
	686			240	173	302
	638			240	124	294
				320	112	297
				400	95	309
				500	90	327

Range of possible elevator configurations for SM200C

Important technical data for the drawn configuration examples are listed in the table below.
The following diagram shows the range of elevator configurations which can be achieved with the ZAtop SM200C elevator machine.
Other elevator configurations, also outside this range, are possible. Flexibility is our business!
Our calculation software ZAlift is available free of charge to you for fast and convenient calculation of your elevator project.

Example configurations

Suspension	No	Max. payload	Speed	Motor type	Traction sheave	Rope	Motor power	Rated current
		kg	m/s		mm	Number x Ø mm	kW	A
1:1	I	300	1.0	SM200.15C	240	6 x 6.5	2.0	8.8
	I	300	1.0	SM200.30C	400	3 x 10	3.4	12.8
	II	450	1.0	SM200.30C	240	8 x 6.5	3.1	11.3
	III	630	1.0	SM200.40C	320	5 x 8	4.0	18.8
	IV	675	1.0	SM200.20C	160	8 x 6.5	4.2	12.9
	V	675	1.6	SM200.20C	160	8 x 6.5	6.7	19.2
	VI	800	1.0	SM200.40C	160	9 x 6.5	4.9	13.8
	VII	800	1.6	SM200.40C	160	9 x 6.5	7.9	21.0
2:1	I	450	1.25	SM200.15C	320	4 x 8	4.2	14.3
	II	630	1.6	SM200.20C	320	4 x 8	7.2	20.8
	III	675	1.0	SM200.20C	320	4 x 8	4.8	14.8
	III	675	1.0	SM200.15C	200 / 210	7 x 6....	4.9	15.4
	IV	675	1.6	SM200.15C	200 / 210	7 x 6....	7.8	22.1
	V	800	1.0	SM200.20C	200	7 x 6....	5.7	16.3
	V	800	1.0	SM200.20C	240	7 x 6....	5.7	17.0
	VI	800	1.6	SM200.30C	320	4 x 8	8.9	22.8
	VII	1000	1.0	SM200.40C	320	6 x 8	7.1	19.9
	VIII	1000	2.5	SM200.40C	320	6 x 8	17.8	41.7
	IX	1200	1.0	SM200.20C	160	6 x 6.5	7.7	23.2
	X	1200	1.6	SM200.30C	200	6 x 6.5	12.2	29.7
	XI	1600	1.0	SM200.40C	240	10 x 6.5	10.4	28.0
	XII	1600	1.6	SM200.40C	240	10 x 6.5	17.1	41.6
	XIII	1800	1.0	SM200.40C	160	10 x 6.5	19.3	45.2